Data Science

Capstone Report - Fall 2024

Predicting Depression Using Resting-State EEG Signals

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supervised by
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Declaration

I declare that this senior capstone was composed entirely by myself with the guidance of my advisor, and that it has not been submitted, in whole or in part, to any other application for a degree. Except where it is acknowledged through reference or citation, the work presented in this capstone is entirely my own.

Preface

I use resting state EEG to classify depression and healthy subjects to help us better understand the neuronal mechanism of depression.

Acknowledgements

I want to thank my parents, Lv Ning and He Wei, who unconditionally supported me throughout my journey and my dog, Duck, who gave me energy during my toughest period in life.
I also have a special thanks to Prof. Xing Tian who supervised this project. He led me to the field of neuroscience when I decided to transform my career from business to neuroscience to uncover the deeper mechanism of human behavior.

Abstract

This study explores the potential of aperiodic neural components, spectral entropy, and functional connectivity as biomarkers for Major Depressive Disorder (MDD) using a resting-state EEG dataset. MDD poses significant diagnostic challenges due to its complex neurophysiological underpinnings. To address these issues, I employed advanced feature extraction, feature selection, and machine learning techniques to identify key predictors, such as the aperiodic knee and offset, which indicate increased low-frequency oscillatory activity and intrinsic neuronal timescale alterations. Spectral entropy analyses revealed heightened randomness in neural activity, particularly in frontal regions, while functional connectivity analyses highlighted disrupted coherence across theta, alpha, and gamma frequency bands, alongside compensatory increases in delta. Using these biomarkers, I developed five binary classification models, including k-NN, SVM, Logistic Regression, Random Forest, and Gradient Boosting, with k-NN and SVM achieving the highest accuracy of 93.3% in 10-fold cross-validation. Future directions include integrating biomarker explanations into user-friendly clinical interfaces and validating findings on diverse populations.

Keywords

NYU Shanghai; Capstone; Data science; Neuroscience; Computational Psychiatry

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1 Introduction

Major Depression Disorder (MDD) is a mental disorder characterized by persistent low mood, loss of pleasure or interests in activities, recurrent thoughts of death, and physical and cognitive impairment [1]-[3]. It is a widespread disease affecting 185 million people globally [4], and it has severe damage to a person's mental and physical wellbeing. MDD patients experience stigma, higher risk of physical comorbidities, such as pain, diabetes, cardiovascular disease, stroke, and Alzheimer's disease, and increased risk of adverse outcome in education, employment, and personal relationships [5]-[10]. 15% of MDD patients die from suicide [11].

The diagnosis of MDD mainly relies on subjective clinical interview and self-questionnaires, and there is no non-invasive and quantitative test available for diagnosis of MDD [12]. Objective biomarkers will not only be helpful in diagnosis, but also help us understand neurophysiological mechanism of MDD. Electroencephalogram (EEG) is a non-invasive, cost-efficient tool to capture the dynamics of brain electric activities with high temporal resolution, and EEG power spectra analysis is a commonly adopted method in the psychiatric disorder studies [13]-[14]. Some EEG power spectra features have been shown to associated with depression, which may provide insights for an objective diagnosis.

The goal of my study is to select promising new EEG signal features that lacks validation from the literature and build prediction models using resting-state EEG data. I aim to validate how efficient the selected features are at predicting depression and how the results related to the neuronal mechanism of MDD. I adopted 5 different binary classifiers to comprehensively compare the efficiency of these models in predicting MDD.

2 Related Work

Resting-state EEG is crucial as it minimizes task-related confounds, reflects spontaneous self-relevant brain activity, and highlights MDD-specific characteristics like increased self-focused thinking and impaired cognitive processing. Classifying depression and healthy subjects using resting-state EEG data is widely studied but still has challenges.

2.1 Features in predicting MDD

3 EEG features which may be associated with MDD are selected from the literature. One promising feature is the aperiodic component of the brain power spectrum, which reflects the baseline of

brain oscillation activities and has captured researchers' attention in recent years. Flatter slopes in aperiodic components during sleep have been observed in MDD patients and are associated with reduced severity and better treatment response [16]-[18]. Additionally, spectral entropy quantifies the complexity and irregularity of EEG signals in the frequency domain, indicating the degree of randomness in brain activity. It has been noted for its potential in capturing cognitive dysregulation in disorders like schizophrenia, making it a candidate biomarker for MDD as well [19]. Moreover, functional connectivity features, such as coherence, measure the consistency of phase relationships between signals in different brain regions over time. Coherence has been shown to reflect the strength and synchronization of functional networks and is one of the most effective predictors of depression in resting-state EEG classification studies [20]. Increased beta frequency band connectivity is already observed in resting-state EEG of MDD patients [21].

2.2 Classification of MDD using resting-state EEG

Resting-state EEG has been widely used for Major Depressive Disorder (MDD) classification through diverse machine learning techniques. Support Vector Machines (SVM) have achieved over 90% accuracy [22], while synchronization likelihood-based SVM reached 98% accuracy [23]. K-nearest neighbors (KNN) with GreedyStepwise feature selection achieved up to 98% accuracy in detecting mild depression [24]. While deep learning has shown promise for automated feature extraction, many studies still prefer traditional approaches due to limited dataset sizes and the high performance already achieved by handcrafted features. Additionally, deep learning models often produce abstract features that lack direct neuroscience significance.

Although there are many established models with high accuracy, this project is still valuable for several reasons. First, the 3 selected features, aperiodic component, spectra entropy, and coherence still lack validation on MDD. Second, generalization of models replies on more datasets that representing difference populations, my models are built on a dataset with MDD medicated outpatients which I will introduce later. Lastly, I adopted 5 different models which provides a comprehensive comparison across model efficiency.

3 Solution

The analysis was based on the 5-min time series data with 19 channels for 64 subjects paired with labels of MDD or HC. I put a lot of effort in the preprocessing step because EEG recording

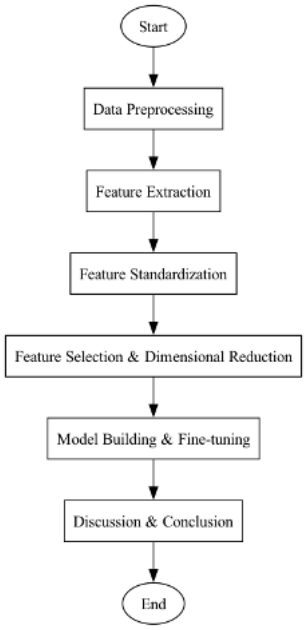

Figure 1: Workflow of the study

often contain noise and artifacts other than brain activities. After that, I extracted features from these time series data with their locational information. Then I conducted features selection and dimensional reduction and build 5 binary classifiers. Besides, I also conducted a Welch's t-test for the features extracted to help understand the importance of these features and the mechanism of depression.

3.1 Dataset

I adopted an open dataset on Figshare containing raw resting-state 5-min EEG recording for 34 clinically diagnosed MDD outpatients and 30 healthy controls (HC) with SSRI's medication treatment [25]. To wash out medication effect, participants had gone through a 2-week no medication period before the EEG recording [26]. The demographic and treatment statistics are shown in the table below. The EEG recording adopted a 19-channel international 10-20 placing standard and a sampling frequency of 256 Hz [26]. The datasets contained both eye open and eye closed condition. I only used the eye closed condition because visual processing is not related to the main symptoms of MDD and eye open condition may introduce more eye movement artifacts.

3.2 EEG data preprocessing

The preprocessing refered to standard procedures in many neuroscience and psychology research. I first truncated all time-series data into 5-min segment. I set montage information based on the

Age	40.3 (±12.9)
Gender	17/17
Pretreatment BDI	20.6 (±8.6)
Pretreatment HADS	10.7 (±2.4)
Post-treatment BDI-II	7.5 (±5.0)
Post-treatment HADS	7.5 (±5.0)
SSRI Treatment	E: 14, F: 9, S: 8, FI: 3

Table 1: Demographic and clinical characteristics of participants

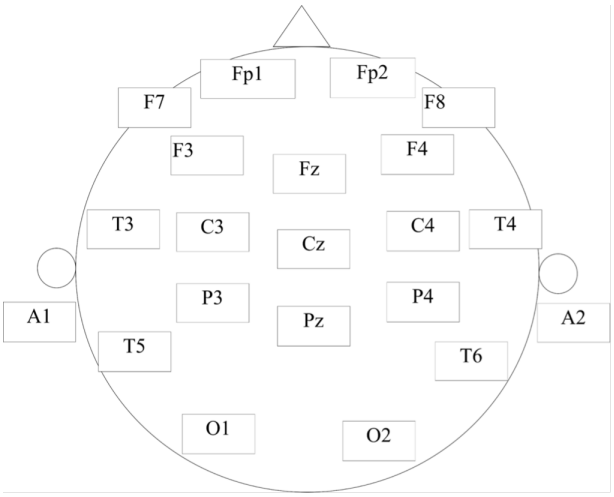

Figure 2: Standard 10-20 EEG electrode placement system used for data acquisition and analysis

international 10-20 standard for each segment and saved them into .fif files. Six subjects were lost because of invalid format. Then I downsampled data to 250 Hz for computational efficiency and help merge with other datasets in future study. Signals other than brain activities might be presented in the whole signals such as muscle artifact, heart rate artifact, and alternating current artifacts. To eliminate this noise, I adopted a 1-45 Hz Finite Impulse Response filter to eliminate components with too high or too low frequency because they are often noisy and contain limited information for brain activities. Bad channels were defined by channels with too flat signals (variance < 0.5) or too noisy signals (variance > 500). Bad channels were rejected and interpolated by the nearby channels. Subjects with over 20% rejected channels were considered bad subjects and rejected. After that, I re-referenced signals to average signals of all channels because EEG signals are measured as differences in potential between electrodes and average reference can help reduce noise across electrodes. Finally, I conducted an independent component analysis and rejected component with extreme amplitude ($>$100) that is out of the range of normal brain activity.

MDD group had a higher reject rate in subject-wise, channel-wise, and IC-wise than HC group,

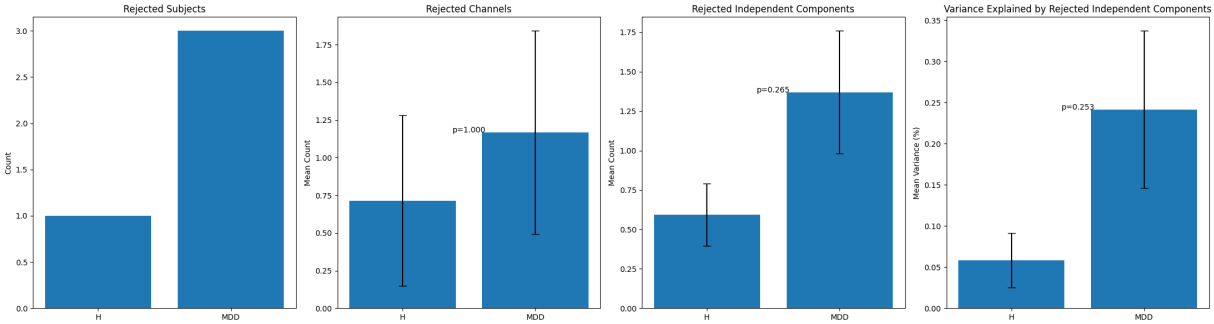

Figure 3: Comparison of rejected subjects, channels, independent components, and variance explained by rejected components between healthy controls (H) and MDD patients, with corresponding p-values indicating statistical significance

but no significant difference between 2 groups was observed.

3.3 EEG feature extraction

Following data preprocessing, the power spectral density (PSD) of the EEG signals was computed using Welch's method, which converted time-series data into the frequency domain to capture the power distribution across frequencies. The FOOOF (Fitting Oscillations and One Over F) module was then applied to decompose the PSD into periodic (oscillatory peaks) and aperiodic (background curve) components. "knee" mode was selected in the FOOOF object because knee makes the curve better capturing low frequency aperiodic components which often has a turning point around 15 Hz. "min_peak_height" is set to 0.1 and "max_n_peak" is set to 7 to ensure only meaningful oscillatory peaks were detected in the PSD while avoiding overfitting or capturing noise. In the end, each channel had 3 aperiodic parameters, offset, knee, and exponent, descripting the aperiodic components of PSD.

The neural power spectrum (NPS) across a range of frequencies (F) can be modeled as:

$$NPS(F) = L(F) + G(F)_n \tag{1}$$

where:

- $L(F)$ represents the aperiodic component.
- $G(F)_n$ represents the periodic component composed of N Gaussian peaks.

The aperiodic component is defined as:

$$L(F) = b - \log(k + F^x) \tag{2}$$

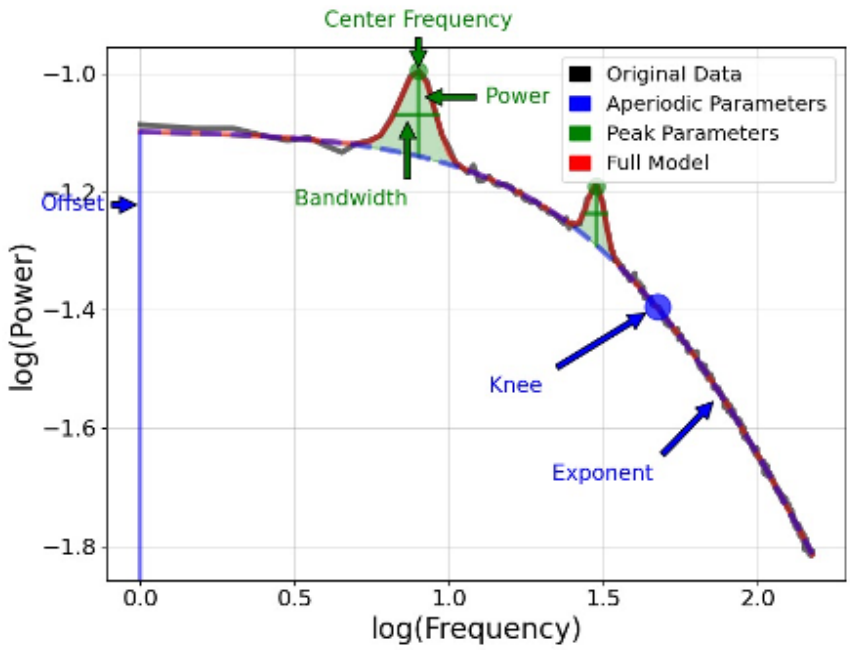

Figure 4: Power spectral density model illustrating key parameters, including aperiodic components (offset, knee, and exponent) and peak features (center frequency, power, and bandwidth), used to characterize neural oscillatory activity

where:

- b: Offset, representing the baseline level of the power spectrum.
- k: Knee, indicating the frequency where the slope changes.
- x: Exponent, which captures the slope of the aperiodic component in the log-log scale.

This component captures the background, aperiodic structure of the neural power spectrum, which generally decreases with increasing frequency.

The periodic component is defined as:

$$G(F)_n = a \cdot \exp\left(-\frac{(F-c)^2}{2 \cdot w^2}\right) \tag{3}$$

where:

- a: Amplitude, representing the height of the Gaussian peak.
- c: Center frequency, where the peak occurs.
- w: Bandwidth, indicating the width of the Gaussian peak.

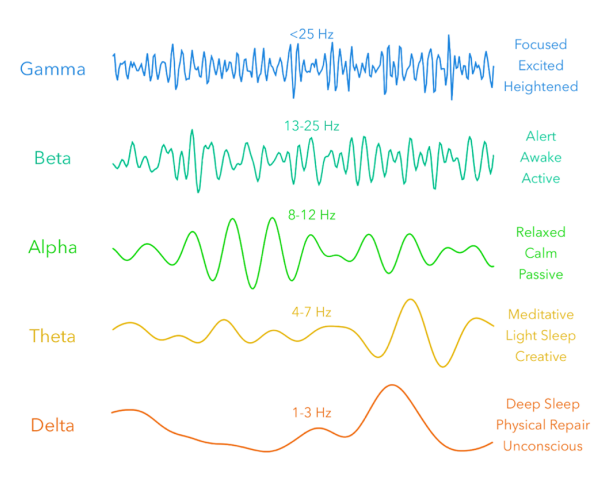

Figure 5: Neural oscillation frequency bands (Gamma, Beta, Alpha, Theta, Delta) with their corresponding frequency ranges and associated cognitive and physiological states

This Gaussian function represents periodic oscillations in the power spectrum, typically capturing distinct rhythmic brain activities at specific frequencies.

Spectral entropy quantifies the complexity of a signal in the frequency domain, with higher values representing more uniform power distribution across frequencies (e.g., noise) and lower values indicating concentrated power in specific rhythmic brain activities. Spectral entropy for each channel was calculated using the "antropy" library's "spectral_entropy" function, which normalizes the power spectral density (PSD) and applies the formula. Each channel is represented by a single scalar value that describes its signal randomness.

$$H = -\sum P(f) \log(P(f)) \tag{4}$$

Signal coherence measures the synchronization between pairs of EEG channels across different frequency bands (delta, theta, alpha, beta, gamma). It was calculated using the "scipy" library's "coherence" function, which computes the cross-power spectral density between two signals and normalizes it by their individual power spectra. Coherence values were averaged within each band, resulting in a set of features capturing functional connectivity for all channel pairs across the frequency bands. These features provide insights into the degree of inter-regional synchronization in brain activity.

$$C_{xy}(f) = \frac{|P_{xy}(f)|^2}{P_{xx}(f) \cdot P_{yy}(f)} \tag{5}$$

where:

- $P_{xy}(f)$: Cross-power spectral density of $x(t)$ and $y(t)$.
- $P_{xx}(f)$: Power spectral density of $x(t)$.
- $P_{yy}(f)$: Power spectral density of $y(t)$.
- $|P_{xy}(f)|^2$: Magnitude squared of the cross-power spectral density.

3.4 Data imputation and normalization

The results consisted of 931 features derived from 19 EEG channels, including aperiodic parameters, spectral entropy, and coherence, measured across predefined frequency bands. Missing and infinite values in the data were imputed using the mean values of the respective group (MDD or healthy control) to ensure consistency. For normalization, the StandardScaler from the sklearn library was applied to transform all features to have a mean of 0 and a standard deviation of 1. This preprocessing step ensures that the features were scaled comparably, facilitating effective performance across machine learning models. The final dataset was clean, normalized, and saved into a pandas datafame for further analysis.

3.5 Feature selection and dimensional reduction

Feature selection and dimensional reduction were conducted to identify key predictors and streamline the dataset for classification. Recursive Feature Elimination (RFE) with a Random Forest classifier was used to rank features based on their predictive importance, selecting the top 25 out of 931 features with a step size of 10. To further reduce dimensionality, Principal Component Analysis (PCA) was applied to these 25 features, where the first principal component captured approximately 79.91% of the variance, and the next two components captured 6.45% and 3.36%, respectively. This process ensured that the dataset retained the most meaningful information while minimizing redundancy, optimizing it for classification tasks.

3.6 Building and fine-tuning binary classifiers

Five models – Logistic Regression (LR), k-NN, Supportive Vector Machine (SVM), Random Forest (RF), and Gradient Boosting (GB) – were trained to classify MDD and HC with the processed features. Hyperparameters were optimized via grid search. 10-fold cross-validation ensured robust evaluation by splitting the small dataset into training (80%) and validation (20%) folds, reducing overfitting and providing reliable matrix.

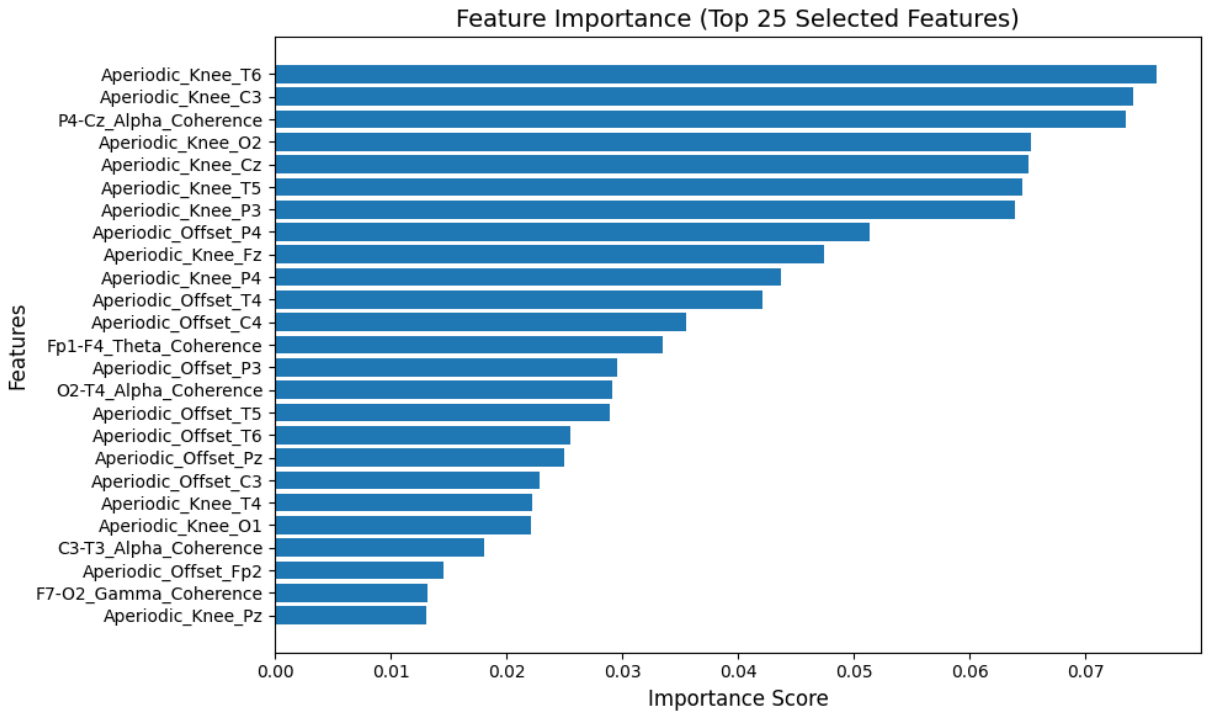

Figure 6: Feature importance scores of the top 25 selected features, highlighting aperiodic parameters (knee and offset) as the most significant predictors for distinguishing MDD from healthy controls

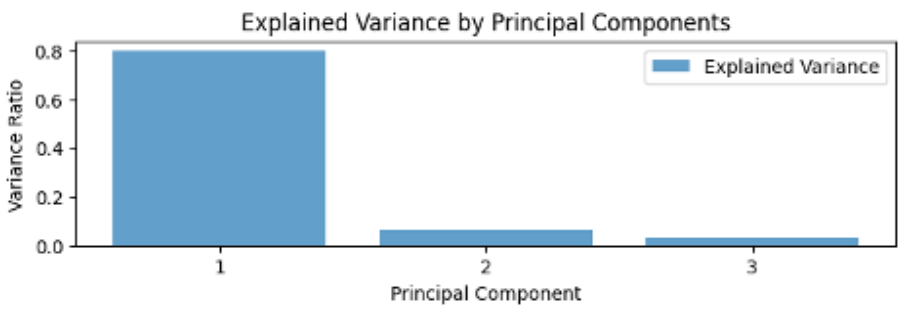

Figure 7: Explained variance ratio by the top three principal components

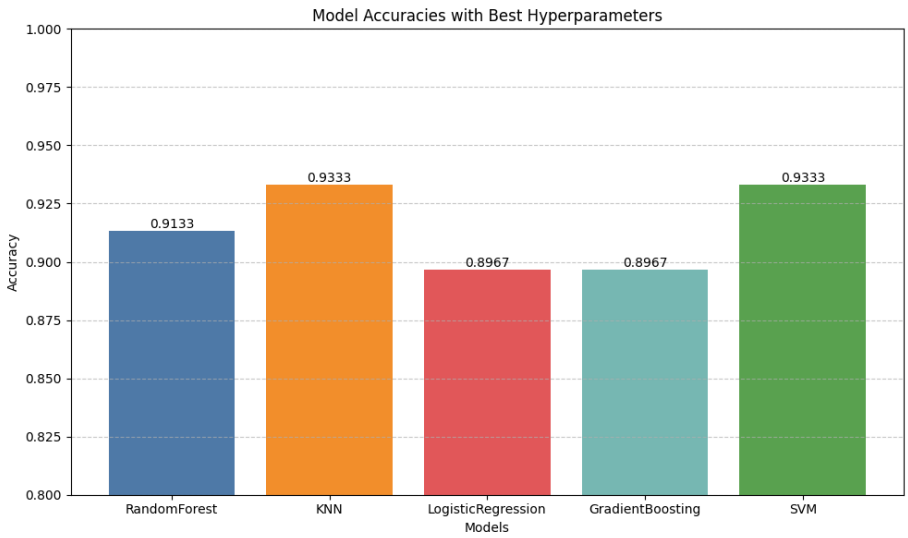

Figure 8: Model accuracies with optimized hyperparameters, average across 10-fold

4 Results

The results demonstrated the effectiveness of the selected models and features in accurately distinguishing between MDD and HC groups, with k-NN and SVM achieving the highest performance metrics, and feature importance analysis highlighting the aperiodic knee and specific brain regions as key predictive elements.

4.1 Model comparison

k-NN and SVM achieved highest mean accuracy (93.33% ± 0.1106) with consistent precision (92.50% ± 0.1106), recall (97.5% ± 0.0750), and F-1 scores (94.64% ± 0.0864) across 10 folds, suggesting reliable performance across different subsets of the data. RF follows closely with 91.33% (± 0.1147) accuracy while LR and GB show slightly lower accuracy (both 0.8967 ± 0.1545). The simple models achieving high accuracy indicates that the data is well processed and structured, and the selected features are reliable in distinguishing MDD and HC.

4.2 Feature importance analysis

Feature importance analysis highlighted the aperiodic knee among the selected 25 features as the most significant predictor, contributing 55.7% of the overall prediction power. This finding suggests that the knee parameter may provide a more robust representation of aperiodic component shifts compared to the exponent (slope), which interestingly does not appear among the top features in this analysis, despite its prominence in many prior studies [16, 17].

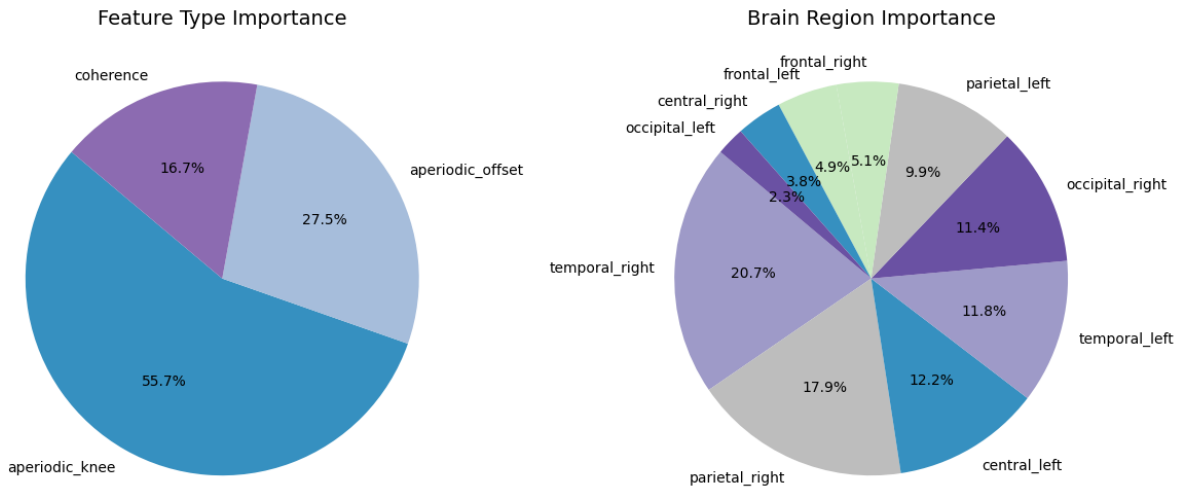

Figure 9: Feature importance distribution derived from a 10-fold cross-validation random forest model applied to the top 25 selected features, showing the dominance of aperiodic knee (55.7%) among feature types, and brain region importance, highlighting the right temporal (20.7%) and right parietal (17.9%) lobes as key contributors

Regarding brain region contributions, the right hemisphere demonstrated greater predictive power, with the right temporal lobe (20.7%) and right parietal lobe (17.9%) emerging as key contributors. These regions outperformed other areas in predicting the studied outcomes, emphasizing their importance in the feature set. Other significant contributions are observed in the left temporal lobe (11.8%) and right occipital lobe (11.4%).

5 Discussion

This study highlights the potentials of aperiodic components, spectral entropy, and functional connectivity as biomarkers for distinguishing individuals with Major Depressive Disorder (MDD). By conducting Welch's t-tests with False Discovery Rate (FDR) corrections for all features, the findings reveal significant alterations in brain functionality across various regions and frequency bands, shedding light on the neural underpinnings of MDD.

5.1 Aperiodic components as a biomarker

Aperiodic features, particularly the knee, emerged as the most significant predictor, suggesting their sensitivity to intrinsic neuronal timescales (INTs). The INT reflects the time over which neurons integrate information, with shorter timescales indicating faster, more transient processing and longer timescales enabling sustained activity [27]. In MDD, an increased knee suggests a shift

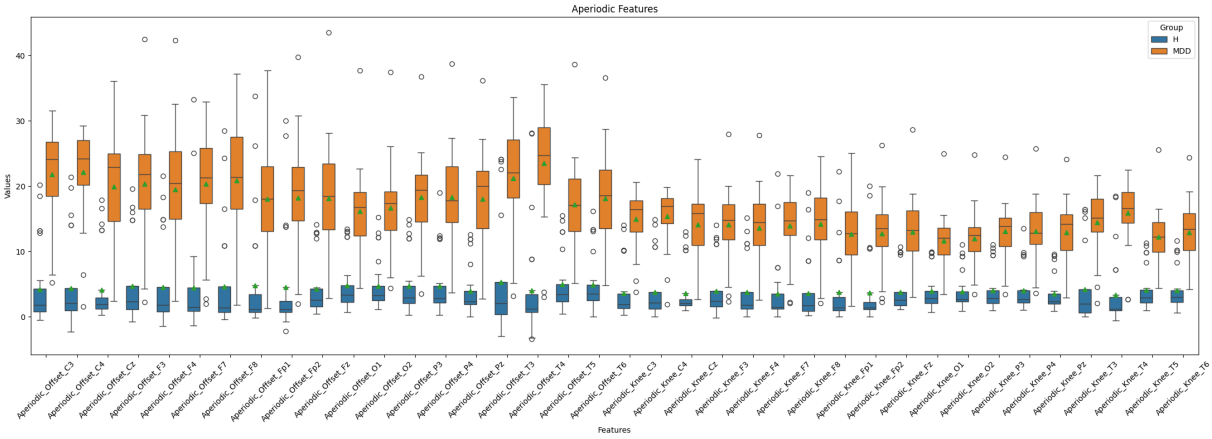

Figure 10: All significant aperiodic features in the all-feature Welch's t-test on MDD and HC groups with FDR correction with 95% confidence level, showing elevated knee and offset values in the MDD group, consistent with increased low-frequency oscillatory activity

toward prolonged INTs, which may reflect enhanced low-frequency neural synchronization and reduced cortical excitability. Such alterations could disrupt dynamic information processing and lead to heightened internal focus or impaired responsiveness to external stimuli. The global increased of offset and knee indicates an elevation of low frequency oscillation in brain's activities in line with previous findings that there are increased theta, alpha, and beta band amplitude in MDD and potentially impairs information processing and promoting maladaptive internal focus in MDD [28]. Additionally, the lack of significance in exponent (slope) features is in line with the feature selection result, which emphasizes the offset and knee together may be better at the capturing the aperiodic component shift in MDD.

5.2 Brain oscillations are more random across frequency

Increased spectral entropy in the MDD group indicates more random and disorganized neural oscillations, suggesting a loss of stability in brain networks [29]. This randomness could impair efficient signal transmission and integration, contributing to cognitive and emotional dysfunction. Notably, the most pronounced entropy changes were observed in the frontal lobe, which plays a critical role in executive functions, emotional regulation, and decision-making [30]. Disruption in these processes aligns with the core symptoms of MDD, such as impaired focus, emotional dysregulation, and difficulty in adapting to environmental demands.

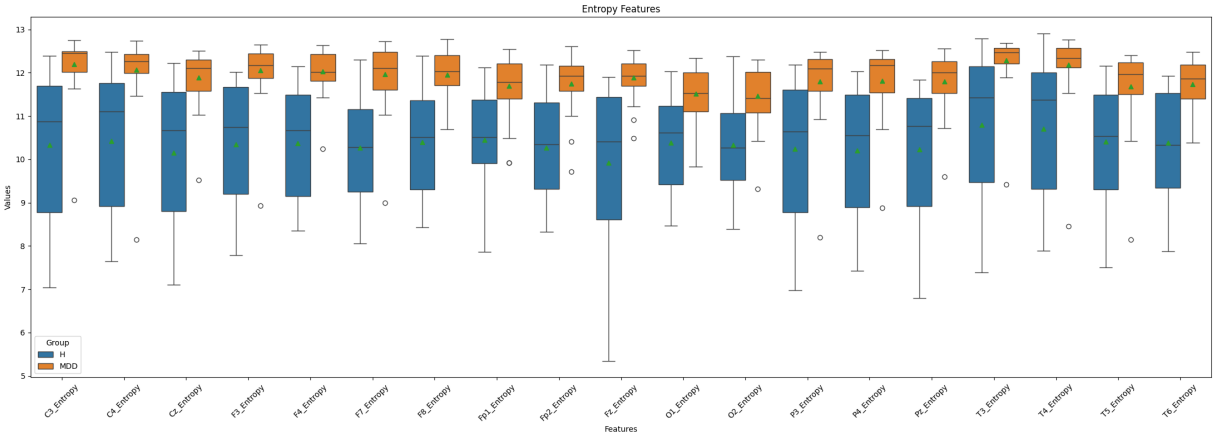

Figure 11: All significant entropy features in the Welch's t-test on MDD and HC groups with FDR correction at a 95% confidence level, showing increased spectral entropy in the MDD group, consistent with disorganized and random neural oscillatory activity

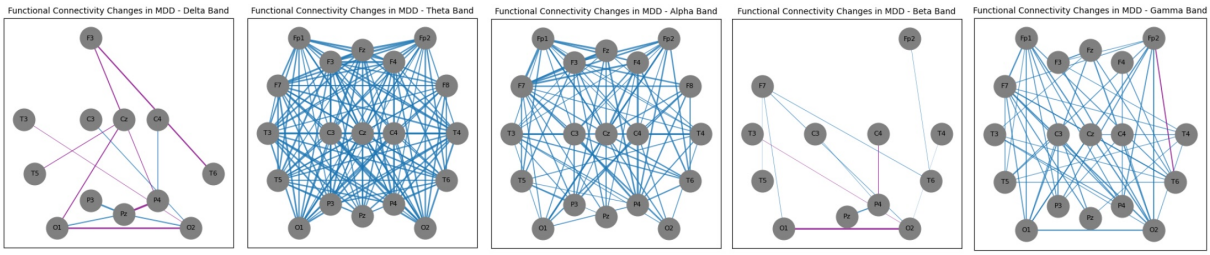

Figure 12: Brain functional connectivity maps across five frequency bands in MDD and HC groups, with blue lines indicating decreased coherence and purple lines indicating increased coherence and line widths represent the magnitude of coherence differences, grey circles indicating electrodes with corresponding position on the brain, and only significant electrode pairs in the all feature t-test are shown

5.3 Disrupted functional connectivity in MDD

Functional connectivity analysis revealed significant abnormalities in coherence across frequency bands. Specifically, reduced theta, alpha, and low gamma coherence suggests impaired synchronization within large-scale networks, likely stemming from disrupted GABAergic inhibitory control [31]. This dysfunction may underlie deficits in memory, emotional processing, and attentional regulation [32]. Conversely, increased delta coherence potentially represents compensatory mechanisms attempting to preserve communication in a dysfunctional network. The overall pattern reflects a compromised balance between neural excitation and inhibition, contributing to the widespread cognitive and emotional challenges observed in MDD.

6 Limitations and Future Directions

This study is limited by the small sample size, which may reduce the generalizability of findings, and the lack of external validation on datasets. The exclusive focus on outpatients receiving SSRIs may also limit the representativeness of the MDD sample. While aperiodic features and coherence metrics provide valuable insights, their specificity to MDD requires further investigation, as they may overlap with other psychiatric or neurological conditions. Future research should explore these features across broader populations and incorporate multimodal data to enhance diagnostic precision.

Additionally, future work should focus on developing a user-friendly interface that integrates many features and highlights positive biomarkers detected in patients. The interface should link each biomarker to concise explanation of its neural significance (e.g., how an increased knee relates to slower intrinsic neuronal timescales), its potential symptom (e.g., attention impairment, suicidal tendency), and treatment response prediction (e.g. how likely patient will response to the transcranial magnetic stimulation treatment). This would enhance clinical utility and improve understanding for both practitioners and patients.

7 Conclusion

This study demonstrates the significance of aperiodic features, spectral entropy, and coherence in understanding and diagnosing MDD. Increased low-frequency oscillatory activity, as indicated by the aperiodic knee and offset, reflects key neural alterations. Spectral entropy highlights neural disorganization, while disrupted coherence across frequency bands underscores widespread network dysfunction in MDD. To further validate these findings, future research should employ experimental causal manipulations, such as TMS or animal models, to establish direct links between these biomarkers and symptom changes. These results provide a strong foundation for developing diagnostic tools and interventions, with a focus on expanding datasets, refining models, and integrating clinically actionable insights.

8 Data and Code Availability

Data is availible at Figshare open access. Code and figures are available at my github.

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